

Orbit Science Academy

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Exam : NEET

Marks :

Duration : 1

Q. 1 Chemistry

Identify the element which can have highest oxidation numbers

- A N
- B O
- C Cl
- D C

Q. 3 Chemistry

Assertion : HClO_4 is stronger acidic than HClO_3 .

Reason : Oxidation state of Cl in HClO_4 is + V II and in HClO_3 + V .

- A If both assertion and reason are true and the reason is the correct explanation of the assertion.
- B If both assertion and reason are true but reason is not the correct explanation of the assertion.
- C If assertion is true but reason is false.
- D If the assertion and reason both are false.

Q. 5 Chemistry

In which reaction there is a change in valency

- A $2\text{NO}_2 \rightarrow \text{N}_2\text{O}_4$
- B $2\text{NO}_2 + \text{H}_2\text{O} \rightarrow \text{HNO}_2 + \text{HNO}_3$
- C $\text{NH}_4\text{OH} \rightarrow \text{NH}_4^+ + \text{OH}^-$
- D $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$

Q. 7 Chemistry

How many moles of $\text{K}_2\text{Cr}_2\text{O}_7$ can be reduced by 1 mole of Sn^{2+}

- A $1/3$
- B $1/6$

Q. 2 Chemistry

When sodium metal is dissolved in liquid ammonia, blue colour solution is formed. The blue colour is due to

- A Solvated Na^+ ions
- B Solvated electrons
- C Solvated NH_2^- ions
- D Solvated protons

Q. 4 Chemistry

In the balanced chemical reaction, $\text{IO}_3^- + a\text{I}^- + b\text{H}^+ \rightarrow c\text{H}_2\text{O} + d\text{I}_2$ a, b, c and d respectively correspond to

- A 5, 6, 3, 3
- B 5, 3, 6, 3
- C 3, 5, 3, 6
- D 5, 6, 5, 5

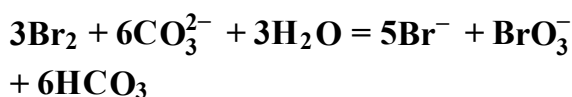
Q. 6 Chemistry

For H_3PO_3 and H_3PO_4 the correct choice is

- A H_3PO_3 is dibasic and reducing
- B H_3PO_3 is dibasic and non-reducing
- C H_3PO_4 is tribasic and reducing
- D H_3PO_3 is tribasic and non-reducing

Q. 8 Chemistry

In the following reaction,



- C 2/3
D 1

Q. 9 Chemistry

The compound $\text{YBa}_2\text{Cu}_3\text{O}_7$ which shows superconductivity has copper in oxidation state Assume that the rare earth element Yttrium is in its usual +3 oxidation state

- A 3/7
B 7/3
C 3
D 7

Q. 11 Chemistry

Which one of the following reactions does not involve either oxidation or reduction

- A $\text{VO}_2^+ \rightarrow \text{V}_2\text{O}_3$
B $\text{Na} \rightarrow \text{Na}^+$
C $\text{CrO}_4^{2-} \rightarrow \text{Cr}_2\text{O}_7^{2-}$
D $\text{Zn}^{2+} \rightarrow \text{Zn}$

Q. 13 Chemistry

Assertion : Fluorine exists only in -1 oxidation state.

Reason : Fluorine has $2s^2 2p^5$ configuration.

- A If both assertion and reason are true and the reason is the correct explanation of the assertion.
B If both assertion and reason are true but reason is not the correct explanation of the assertion.
C If assertion is true but reason is false.
D If the assertion and reason both are false.

Q. 15 Chemistry

The oxidation number of sulphur in S_8 , S_2F_2 , H_2S respectively, are

- A 0, +1 and -2
B +2, +1 and -2

- A Bromine is oxidised and carbonate is reduced
B Bromine is reduced and water is oxidised
C Bromine is neither reduced nor oxidised
D Bromine is both reduced and oxidised

Q. 10 Chemistry

In alkaline medium ClO_2 oxidize H_2O_2 in O_2 and reduced itself in Cl^- then how many mole of H_2O_2 will oxidize by one mole of ClO_2

- A 1.0
B 1.5
C 2.5
D 3.5

Q. 12 Chemistry

What is the oxidation number of Co in $[\text{Co}(\text{NH}_3)_4\text{ClNO}_2]$

- A +2
B +3
C +4
D +5

Q. 14 Chemistry

M^{+3} ion loses $3e^-$. Its oxidation number will be

- A 0
B +3
C +6
D -3

Q. 16 Chemistry

In a reaction between zinc and iodine, in which zinc iodide is formed, what is being oxidised

- A Zinc ions
B Iodide ions

- c 0, + 1 and + 2
 d - 2, + 1 and - 2

- c Zinc atom
 d Iodine

Q. 18 Chemistry

Assertion : Stannous chloride is powerful oxidising agent which oxidises mercuric chloride to mercury.

Reason : Stannous chloride gives grey precipitate with mercuric chloride, but stannic chloride does not do so.

- A If both assertion and reason are true and the reason is the correct explanation of the assertion.
 B If both assertion and reason are true but reason is not the correct explanation of the assertion.
 C If assertion is true but reason is false.
 D If assertion is false but reason is true.

Q. 20 Chemistry

What is the equivalent mass of IO_4^- when it is converted into I_2 in acid medium

- A M/6
 B M/7
 C M/5
 D M/4

Q. 17 Chemistry

Oxidation state of oxygen atom in potassium superoxide is

- A 0
 B - 1
 C $-\frac{1}{2}$
 D - 2

Q. 19 Chemistry

In the reaction $\text{I}_2 + 2\text{S}_2\text{O}_3^{2-} \rightarrow 2\text{I}^- + \text{S}_4\text{O}_6^{2-}$ equivalent weight of iodine will be equal to

- A 1/2 of molecular weight
 B Molecular weight
 C 1/4 of molecular weight
 D None

Q. 21 Chemistry

Match List I with List II and select the correct answer using the codes given below the lists

List I (Compound)	List II
(A) NO_2	(1) + 5
(B) HNO	(2) - 3
(C) NH_3	(3) + 4
(D) N_2O_5	(4) + 1

Codes :

- A A B C D
2 3 4 1
B A B C D
3 1 2 4
C A B C D
3 4 2 1
D A B C D
2 3 1 4

Q. 23 Chemistry

The equivalent weight of phosphoric acid (H_3PO_4) in the reaction $\text{NaOH} + \text{H}_3\text{PO}_4 \rightarrow \text{NaH}_2\text{PO}_4 + \text{H}_2\text{O}$ is

- A 25
B 49
C 59
D 98

Q. 25 Chemistry

The oxidation number of nitrogen in NH_2OH is

- A + 1
B - 1
C - 3
D - 2

Q. 27 Chemistry

The strongest reducing agent is

- A HNO_2
B H_2S
C H_2SO_3
D SnCl_2

Q. 29 Chemistry

Which one of the following reactions is not an example of redox reaction

Q. 22 Chemistry

The conductivity of a saturated solution of BaSO_4 is $3.06 \times 10^{-6} \text{ ohm}^{-1} \text{ cm}^{-1}$ and its equivalent conductance is $1.53 \text{ ohm}^{-1} \text{ cm}^{-1} \text{ equivalent}^{-1}$. The K_{sp} of the BaSO_4 will be

- A 4×10^{-12}
B 2.5×10^{-9}
C 2.5×10^{-13}
D 4×10^{-6}

Q. 24 Chemistry

Oxidation number of As atoms in H_3AsO_4 is

- A - 3
B + 4
C + 6
D + 5

Q. 26 Chemistry

Several blocks of magnesium are fixed to the bottom of a ship to

- A Keep away the sharks
B Make the ship lighter
C Prevent action of water and salt
D Prevent puncturing by under-sea rocks

Q. 28 Chemistry

The oxidation number of N in NH_4Cl is

- A + 5
B + 3
C - 5
D - 3

Q. 30 Chemistry

The oxidation state of I in IPO_4 is

- A +1
B +3

- A $\text{Cl}_2 + 2\text{H}_2\text{O} + \text{SO}_2 \rightarrow 4\text{H}^+ + \text{SO}_4^{2-} + 2\text{Cl}^-$
 B $\text{Cu}^{++} + \text{Zn} \rightarrow \text{Zn}^{++} + \text{Cu}$
 C $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
 D $\text{HCl} + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{Cl}^-$

Q. 31 Chemistry

In which of the following reactions H_2O_2 is a reducing agent

- A $2\text{FeCl}_2 + 2\text{HCl} + \text{H}_2\text{O}_2 \rightarrow 2\text{FeCl}_3 + 2\text{H}_2\text{O}$
 B $\text{Cl}_2 + \text{H}_2\text{O}_2 \rightarrow 2\text{HCl} + \text{O}_2$
 C $2\text{HI} + \text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{I}_2$
 D $\text{H}_2\text{SO}_3 + \text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{SO}_4 + \text{H}_2\text{O}$

Q. 33 Chemistry

The oxidation number of nitrogen in NO_3^- is

- A - 1
 B + 2
 C + 3
 D + 5

Q. 35 Chemistry

The oxidation number of Fe and S in iron pyrites are

- A 4, - 2
 B 2, - 1
 C 3, - 1.5
 D 3, - 1

Q. 37 Chemistry

In which of the following compounds transition metal is in oxidation state zero

- A $[\text{Co}(\text{NH}_3)_6]\text{Cl}_2$
 B $[\text{Fe}(\text{H}_2\text{O})_6]\text{SO}_4$
 C $[\text{Ni}(\text{CO})_4]$
 D $[\text{Fe}(\text{H}_2\text{O})_3](\text{OH})_2$

Q. 39 Chemistry

Which of the following statements is correct

- A Hydrogen has oxidation number - 1 and + 1

- C +5
 D +7

Q. 32 Chemistry

Which of the following is not a reducing agent

- A NaNO_2
 B NaNO_3
 C HI
 D SnCl_2

Q. 34 Chemistry

If 1.2 g of metal displace 1.12 litre hydrogen at normal temperature and pressure, equivalent weight of metal would be

- A 24
 B 12
 C $1.2 \div 11.2$
 D 1.2×11.2

Q. 36 Chemistry

The compound which could not act both as oxidising as well as reducing agent is

- A SO_2
 B MnO_2
 C Al_2O_3
 D CrO

Q. 38 Chemistry

Of all the three common mineral acids, only sulphuric acid is found to be suitable for making the solution acidic because

- A It does not react with KMnO_4 or the reducing agent
 B Hydrochloric acid reacts with KMnO_4
 C Nitric acid is an oxidising agent which reacts with reducing agent
 D All of the above are correct

Q. 40 Chemistry

Oxidation number of P in $\text{Mg}_2\text{P}_2\text{O}_7$ is

- A + 3
 B + 2

- B** Hydrogen has same electronegativity as halogens
- C** Hydrogen will not be liberated at anode
- D** Hydrogen has same ionization potential as alkali metals

- C** + 5
- D** - 3

Q. 41 Chemistry

Oxidation number of cobalt in $K[Co(CO)_4]$ is

- A** + 1
- B** + 3
- C** - 1
- D** - 3

Q. 43 Chemistry

Oxidation number of Fe in $K_3[Fe(CN)_6]$ is

- A** + 2
- B** + 3
- C** + 1
- D** + 4

Q. 45 Chemistry

The number of electrons involved in the reduction of $Cr_2O_7^{2-}$ in acidic solution to Cr^{3+} is

- A** 0
- B** 2
- C** 3
- D** 5

Q. 42 Chemistry

Reduction involves

- A** Loss of electrons
- B** Gain of electrons
- C** Increase in the valency of positive part
- D** Decrease in the valency of negative part

Q. 44 Chemistry

$2MnO_4^- + 5H_2O_2 + 6H^+ \rightarrow 2Z + 5O_2 + 8H_2O$. In this reaction Z is

- A** Mn^{+2}
- B** Mn^{+4}
- C** MnO_2
- D** Mn